

Jayson C. Sepeda

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Computer Science graduate with hands-on experience in Machine Learning, UI/UX, and full-stack web/app development. Recognized with Best Abstract Award at an international research conference. Strong foundation in GRU-based Speech Emotion Recognition, React and Flask Web Development, AppSheet automation, and App Development with UI/UX prototyping using Flutter and Figma.

Skills

- **Core Skills:** Machine Learning, Neural Networks, Deep Learning, Software Development, Cybersecurity Basics
- **Programming:** Python, Java, C++, C, Dart, PHP, HTML, CSS, JavaScript
- **Web/App Development:** React, Flask, Django, Laravel, Livewire, AppSheet, Looker Studio, MySQL, Tailwind CSS, Flutter
- **UI/UX & Tools:** Figma, Tkinter, Git, Github Google Apps Script
- **Soft Skills:** Analytical Thinking, Attention to Detail, Leadership, Time Management, Team Player

Key Achievements & Published Works

- **Best Abstract Award Winner** - 1st ASEAN Multidisciplinary Research Conference 2024 (International Society of Transdisciplinary Researchers, Educators, and Leaders Inc.)
- **Published Author** - *Enhancement of Gated Recurrent Unit for Speech Emotion Recognition in the Implementation of Voice-Based Danger Recognition System* (Published at United International Journal for Research & Technology, Indexed on Google Scholar and ResearchGate)
 - DOI: <http://dx.doi.org/10.13140/RG.2.2.21381.26085>

Experience

Application Developer & IT Intern | Third Generation Holdings Corporation July 04, 2024 – August 23, 2024

- Created an internal document request, retrieval, and form creation application using Google AppSheet, significantly reducing manual labor and streamlining operational processes.
- Spearheaded end-to-end application testing by distributing a test version across all company departments to validate functionality, gather feedback, and ensure alignment with operational needs.
- Designed and deployed dynamic dashboards and reports using AppSheet and Looker Studio, improving internal reporting efficiency.
- Automated document generation workflows via Google Sheets and Apps Script, reducing manual work by ~50%.
- Customized forms and backend workflows to streamline data input and retrieval processes.
- Provided IT support for hardware/software troubleshooting and issue resolution across departments.
- Conducted system evaluations and submitted structured defect reports to improve internal tools.

Projects

Gated Recurrent Unit (GRU) Voice-Based Danger Recognition System | Lead Programmer

May 2024 – May 2025

Programming Language, Software & Framework: Python, Kaggle, Flask, React-Vite

- Built a GRU-based neural network to classify emotional states from speech signals (sad, happy, angry, fear, neutral, and disgust) using a custom Kaggle dataset.
- Identified and prepared the appropriate dataset, handling data preprocessing and formatting to ensure compatibility with the model pipeline.
- Improved model performance by addressing three core issues in the base GRU model and applying targeted enhancements to each.
- Created a full-stack web system with a React Vite frontend and Flask backend capable of accepting live microphone recordings or uploaded .wav files for emotion classification.
- Engineered backend logic to detect "fear" emotion and trigger a danger recognition response, enabling real-time alert capability.
- Designed and implemented the preprocessing pipeline, data splitting strategy, and model evaluation process to ensure training efficiency and reliable metrics.

- Authored the research manuscript and prepared the system for academic publication and thesis presentation.

KapehaNila | Project Lead & UI/UX

May 2025

Software: Figma

- Led a cross-functional team in the design and planning of a proposed café web app, coordinating all aspects of design, functionality, and deliverables.
- Facilitated regular team meetings to align on milestones, discuss feature implementations, and ensure team productivity.
- Designed the full UI for both mobile and desktop platforms using Figma, including wireframes, low-fidelity, and high-fidelity layouts.
- Built a fully interactive prototype for application review.
- Ensured on-time delivery of all project outputs including the manuscript, design assets, and supporting documentation.

PLM Archive System | Programmer and Documentation Lead

September 2024 – December 2024

Language & Framework: Python, Django, MySQL

- Developed core backend functionalities using Django and MySQL for a centralized web-based university thesis repository system.
- Implemented dynamic user roles and authentication logic (Admin, Faculty, and Student) to manage access, submission, and approval workflows.
- Built and styled responsive frontend pages using HTML, CSS, JavaScript, and Bootstrap, improving usability and navigation.
- Created reusable UI components and integrated backend APIs for seamless data handling and interaction.
- Used Git and GitHub for collaborative version control, managing feature branches and pull requests within a team environment.
- Co-authored system documentation, including a user manual, deployment guide, project brochure, and presentation materials.

PrismViz | UI/UX & Q.A Tester

September 2024 – December 2024

Software: Figma

- Led team meetings to align on project goals, gather feedback, and ensure timely delivery of design milestones and documentation.
- Designed a responsive, user-friendly interface for an online data cleaning platform aimed at simplifying complex data management workflows.
- Created detailed wireframes as well as low- and high-fidelity UI designs for both desktop and mobile experiences using Figma.
- Built interactive design prototypes to support usability testing, stakeholder validation, and iterative design improvements.
- Spearheaded quality assurance (QA) testing of the system UI to identify bugs, visual inconsistencies, and usability issues across devices.
- Delivered comprehensive project documentation, including the user manual, promotional brochure, and final project manuscript.

Mobile Application Development (Renacimiento App & PLM App) | UI/UX & Lead Frontend Developer

March 2024 – June 2024

Software Development Kit & Programming Language: Flutter & Dart

- Led the UI/UX design process, creating detailed wireframes, low-fidelity, and high-fidelity designs in Figma.
- Spearheaded frontend development using Flutter and Dart, translating Figma designs into fully functional, responsive mobile interfaces.
- Facilitated regular team meetings and client check-ins to align on design direction, gather feedback, and track deliverables against deadlines.
- Developed modular and reusable UI components to ensure maintainability, design consistency, and ease of integration with backend services.
- Structured codebase for seamless collaboration with backend developers, enabling efficient full-stack integration.
- Participated in beta testing and QA, helping to identify, document, and resolve bugs and UI/UX inconsistencies prior to final release.
- Contributed to project documentation, including user manuals, promotional brochures, and system flow guides.

Compiler Development | UI/UX & Support Programmer

September 2023 – June 2024

Programming Language & Software: Python, Tkinter, Figma

- Designed a custom programming language and led the creation of its Context-Free Grammar (CFG), serving as the foundational structure for syntax parsing and semantic analysis.
- Developed core compiler components including syntax analyzer and semantic output generator using Python.
- Created reusable functions to streamline lexical and syntax analysis processes, improving modularity and code clarity.
- Designed the compiler's graphical interface using Tkinter and prototyped UI flow with Figma.
- Authored technical documentation detailing the grammar rules, First-Follow-Predict (FFP) sets, and testing scripts.
- Participated in comprehensive testing and debugging phases to ensure correct and efficient code generation.

Academic Management System Web Development | UI/UX

September 2023 – June 2024

Tech Stack & Software: Tailwind CSS, Alpine, Laravel, Livewire, Figma

- Solely designed the full user interface for the Academic Management System (AMS), a centralized platform comprising 7 interconnected subsystems, producing over 100–120 Figma frames covering both low- and high-fidelity prototypes.
- Ensured all designs were aligned with business logic and technically feasible by collaborating closely with the developers and the business analyst throughout the project lifecycle.
- Actively led team discussions on process flow, feature refinement, and usability to maintain consistency and smooth integration across modules.
- Coordinated with the development team to translate UI components into code using Tailwind CSS, Laravel, Livewire, and Alpine.js.
- Maintained design quality by incorporating iterative feedback and gaining approval from stakeholders at each critical milestone.
- Authored complete system documentation, including user manuals and brochures, for stakeholder presentations.

Licenses and Certifications

Artificial Intelligence Fundamentals

July 2025

IBM SkillsBuild

Skills: Python Programming

Python Essentials 1

July 2025

CISCO Networking Academy and Python Institute

Skills: Python Programming

UIJRT: Certificate of Publication

December 2024

United International Journal for Research & Technology

Skills: Machine Learning, Neural Networks, Deep Learning

ISTREL: Certificate of Presentation and Best Abstract Award

December 2024

International Society of Transdisciplinary Researchers, Educators, and Leaders Inc.

Skills: Machine Learning, Neural Networks, Deep Learning

Glitch in the Matrix: Unravel the Secrets of Software Quality Assurance

April 2024

Pamantasan ng Lungsod ng Maynila

Skills: Software Quality Assurance

U & I + X: Breaking Up with Bad Design

November 2023

Pamantasan ng Lungsod ng Maynila

Skills: Essentials of UI/UX Design

Introduction to Cybersecurity

March 2023

CISCO Networking Academy

Skills: Explain the most common Cyber Threats, Attacks, and Vulnerabilities.

Cybersecurity Essentials

April 2023

CISCO Networking Academy

Skills: Describe the procedures used by cybercriminals and explain the purpose of laws related to cybersecurity.

Professional Reference

Bren Alfred España

Pamantasan ng Lungsod ng Maynila

- baespana24@gmail.com
- 09171388873
- Lecturer III / Professor
- Best time to call 3pm onwards

Education

Pamantasan ng Lungsod ng Maynila

June 2021 - Present

Bachelor of Science in Computer Science | GWA: 1.44